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Rubisco is often used as a model for protein folding studies. On their own, rubisco monomers typically cannot fold to completion and instead are trapped in a form known as a "kinetic intermediate." These kinetically trapped monomers cannot dimerize correctly and are prone to aggregation.

Scientists interested in studying protein folding dynamics labeled the rubisco kinetic intermediate with a set of fluorophores known as a FRET (fluorescence resonance energy transfer) pair, with one fluorophore at the N-terminus and the other at the C-terminus. A FRET signal occurs when the emission spectrum of one fluorophore (the donor) overlaps with the excitation spectrum of the other (the acceptor). When the two fluorophores are sufficiently close to each other, the donor can transfer energy from the light it absorbs directly to the acceptor, causing the acceptor to emit light after the donor absorbs light.

The GroEL/ES complex is a chaperone protein that binds and releases misfolded proteins, hydrolyzing ATP each time the protein is released. A misfolded protein may undergo multiple rounds of binding and release before adopting its correct conformation, or it may never fold correctly and instead be targeted for destruction (Figure 1).

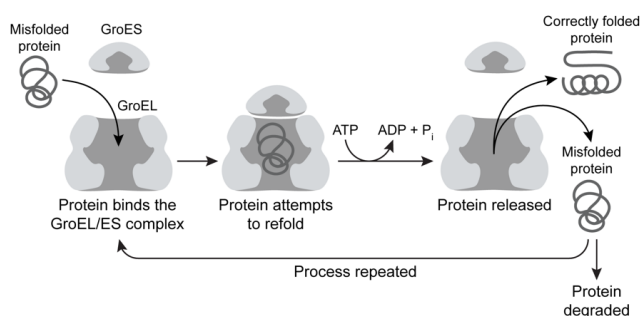


Figure 1 Representation of GroEL/ES-mediated protein folding

Researchers diluted the labeled kinetic intermediate into a refolding buffer in the presence or absence of 200 nM GroEL/ES and then monitored the FRET signal over time. They observed that under the conditions used, rubisco folding was

first-order with respect to [unfolded rubisco] when GroEL/ES was present. The results are shown in Figure 2.

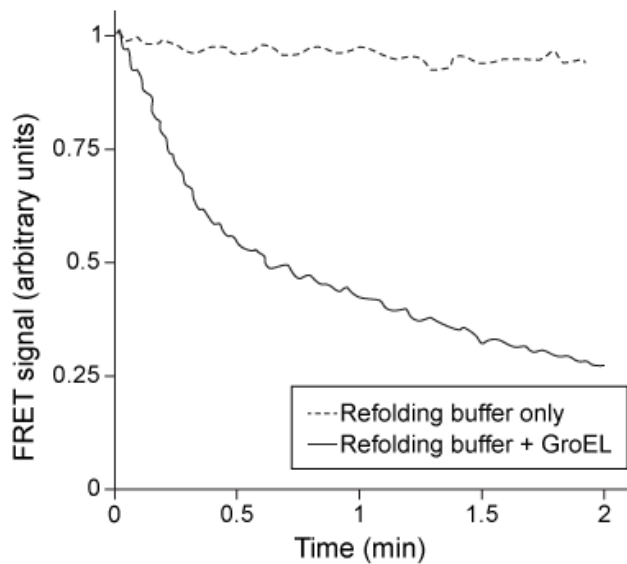


Figure 2 FRET signal of labeled rubisco with and without GroEL

Based on the molecular weight of a rubisco monomer, approximately how many amino acids are present in a rubisco dimer?

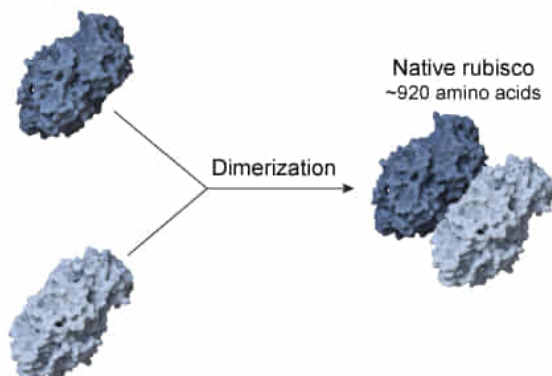
- ☐ A. 460 [20%]
- ☐ B. 690 [11%]
- ✓ ☒ C. 920 [54%]
- ☐ D. 5100 [13%]

Correct

54% answered correctly

Explanation:

Rubisco monomers
51 kDa $\approx$ 460 amino acids each



The **native form of a protein** is the form in which the protein is **biologically active**, and it may consist of **one or more polypeptide chains**. The exact molecular weight of each chain is determined by the primary structure (ie, the sequence of amino acids that makes up the chain) with each amino acid contributing its own molecular weight to the total. The 20 standard amino acids have an **average molecular weight of 110 Da**, so the approximate number of amino acids in a protein can be estimated by dividing its molecular weight in daltons by 110.

The passage states that the functional (native) form of rubisco is a dimer in which each subunit contributes 51 kDa (51,000 Da) of mass. This mass can be converted to the approximate number of amino acids (AA) by

$$\frac{51,000 \text{ Da}}{110 \text{ Da/AA}} = 463 \text{ AA} \approx 460 \text{ AA}$$

However, this is the number of amino acids in a rubisco *monomer* (**Choice A**). Because the native form of rubisco is a homodimer (two identical subunits), the number of amino acids in the monomer must be doubled. Native, dimeric rubisco contains **~920 amino acids**.

(**Choice B**) 690 amino acids is the average of the amino acids in one monomer and one dimer, or 1.5 rubisco subunits. Such a combination does not exist in nature.

(**Choice D**) Dividing the molecular weight of a rubisco monomer by 10 instead of 110 yields 5100 amino acids. A protein with 5100 amino acids would have a mass of approximately 560 kDa, not 51 kDa.

Educational objective:

The native form of a protein is the form in which it is biologically functional. It may consist of one or multiple polypeptide chains. The relationship between the number of amino acids in a protein and its molecular weight can be determined by the fact that, on average, one amino acid contributes 110 Da of mass to the protein.

Time spent: 0 sec

Subject:
Biochemistry

QID: 401134

System:
Amino Acids and Proteins

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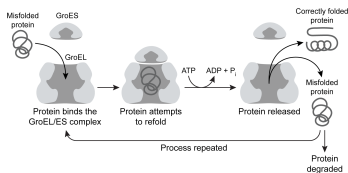


Figure 1 Representation of GroEL/ES-mediated protein folding

Researchers diluted the labeled kinetic intermediate into a refolding buffer in the presence or absence of 200 nM GroEL/ES and then monitored the FRET signal over time. They observed that under the conditions used, rubisco folding was first-order with respect to [unfolded rubisco] when GroEL/ES was present. The results are shown in Figure 2.

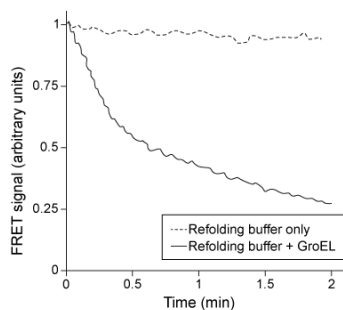


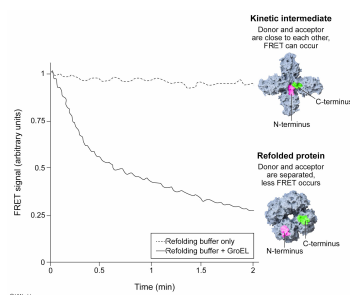
Figure 2 FRET signal of labeled rubisco with and without GroEL

Which of the following conclusions can be drawn based on the data from Figure 2?

- ☐ A. GroEL/ES absorbs light and transfers energy to the FRET donor. [13%]
- ☒ B. GroEL/ES separates the termini of the kinetic intermediate. [61%]
- ☐ C. GroEL/ES and rubisco have overlapping emission and excitation spectra, respectively. [8%]
- ☐ D. GroEL/ES decreases the distance between the termini of rubisco. [16%]

Correct
61% answered correctly

Explanation:



Proteins fold into **three-dimensional structures**. In the folded form, amino acids that are **far apart in the primary structure** may be **close together in space**. Different conformations of the protein may result in different distances between two amino acids. The proximity of amino acids to each other in a folded protein can be probed by techniques such as FRET.

The passage states that a FRET signal occurs when two fluorophores with overlapping excitation and emission spectra (a FRET pair) are near each other. It also states that the N- and C-termini of rubisco were labeled with a FRET pair. Figure 2 shows that in the absence of GroEL/ES, a

relatively strong FRET signal is observed and it remains over the course of the experiment, suggesting that the rubisco kinetic intermediate must be folded such that the termini are near each other.

In the presence of GroEL/ES, the FRET signal is present initially, which is consistent with the fact that initially rubisco was in the kinetic intermediate form. However, over the course of the experiment the **signal decreases**. This indicates that the **fluorophores are being separated** from each other, and therefore GroEL/ES causes the termini of the kinetic intermediate to separate.

(Choice A) The FRET donor absorbs light and transfers the energy to the acceptor. GroEL does not transfer energy to the FRET donor.

(Choice C) The GroEL/ES complex is not labeled with either part of the FRET pair. Therefore, it does not have overlapping excitation or emission spectra with rubisco.

(Choice D) GroEL/ES *increases* the distance between rubisco termini, causing a decrease in the FRET signal.

Educational objective:

Proteins fold into three-dimensional structures. Amino acids that are distant from each other in the linear sequence (primary structure) may be far from or near to each other in space in the protein's folded tertiary and/or quaternary structure. The distance between amino acids may vary as the protein folds.

Time spent: 0 sec
Subject:
Biochemistry

QID: 401135
System:
Amino Acids and Proteins

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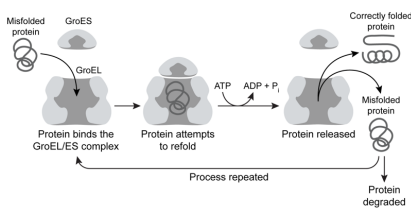


Figure 1 Representation of GroEL/ES-mediated protein folding

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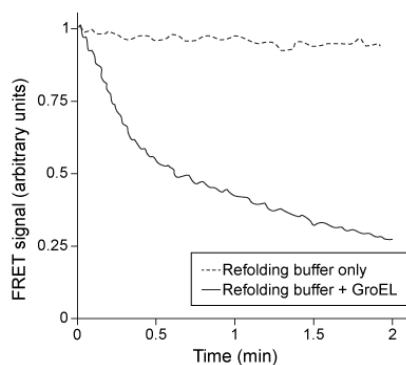


Figure 2 FRET signal of labeled rubisco with and without GroEL

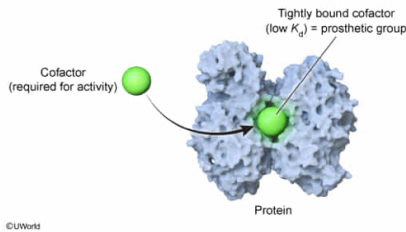
Which of the following is true concerning magnesium in rubisco?

- I. The rubisco- Mg^{2+} interaction has a small K_d .
 - II. Mg^{2+} is a cofactor of rubisco.
 - III. Mg^{2+} activates rubisco by binding at an allosteric site.
- ☐ A. I only [2%]
☒ B. I and II only [73%]
☐ C. II and III only [12%]
☐ D. I, II, and III [10%]

Correct

73% answered correctly

Explanation:



Some proteins, including some enzymes, can function only with the help of non-amino acid groups called **cofactors**. These groups may be organic (in which they are often called **coenzymes**) or inorganic, and they may be tightly or loosely bound to the proteins they help. **Tightly bound cofactors** are called **prosthetic groups**.

The passage states that a magnesium ion (Mg^{2+}) is required on each subunit for rubisco activity. Therefore, Mg^{2+} is a cofactor of rubisco (**Number II**). The passage also states that Mg^{2+} is a prosthetic group, meaning that it is bound tightly to rubisco. The dissociation constant K_d measures the tendency of a protein-ligand complex (ie, rubisco and magnesium) to separate into its components. A **small K_d** indicates a low tendency to dissociate and therefore indicates the tight binding expected of a prosthetic group (**Number I**).

(**Number III**) **Allosteric effectors** bind proteins at positions other than the active site and either increase (activators) or decrease (inhibitors) their activity. The passage says that magnesium ions bind rubisco at the active site; therefore, although magnesium activates rubisco, it does not do so allosterically.

Educational objective:

Some proteins require cofactors (non-amino acid groups) to function. Cofactors may be organic (coenzymes) or inorganic, and they may bind their proteins tightly or loosely. Tightly bound cofactors are called prosthetic groups.

Time spent: 0 sec

Subject:
Biochemistry

QID: 401136

System:
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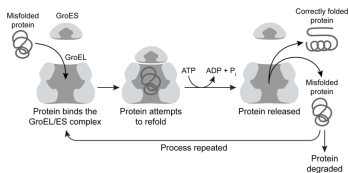


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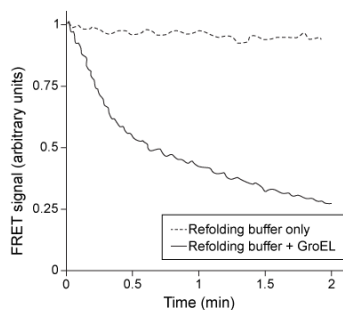


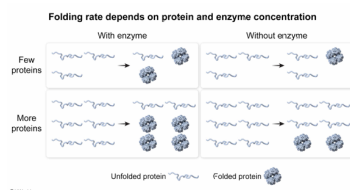
Figure 2 FRET signal of labeled rubisco with and without GroEL

What is the initial rate of rubisco folding in the presence of GroEL/ES, given a rate constant $k_f = 2 \times 10^{-1} \text{ s}^{-1}$?

- ☐ A. 40 s^{-1} [24%]
- ☐ B. 20 s/nM [22%]
- ☐ C. 10 nM [8%]
- ☒ D. There is insufficient information to determine the rate. [44%]

Correct
44% answered correctly

Explanation:



The rate of a process is determined by the concentrations of each reactant raised to an experimentally determined power n . This number is then multiplied by a rate constant that is specific to the reaction. In this case, the **rate of rubisco folding** is dependent upon the **concentration of unfolded protein** raised to the first power (because the passage states that the reaction is **first-order**) and the **rate constant k_f** :

$$\text{Rate} = k_f [\text{unfolded rubisco}]$$

The higher the concentration of unfolded rubisco, the greater the probability that at least some proteins will fold and the faster the process will occur.

The passage states that the experiment is carried out with 200 nM GroEL/ES. As a chaperone protein, GroEL/ES affects the rate of protein folding

by increasing the rate constant of folding, in this case yielding a rate constant of $2 \times 10^{-1} \text{ s}^{-1}$. However, the passage does *not* provide the concentration of rubisco, which must be multiplied by the rate constant ($2 \times 10^{-1} \text{ s}^{-1}$) to determine the overall rate of folding. Because the reaction is first-order with respect to unfolded rubisco, but the concentration of unfolded rubisco is not given, the rate of folding **cannot be determined**.

(Choice A) This option incorrectly uses the concentration of *GroEL/ES* rather than rubisco as the concentration of substrate, and it assumes that the reaction is first order with respect to the *chaperone*. In addition, the units for a reaction rate should be nM/s, not s^{-1} .

(Choice B) In addition to the fact that there is not sufficient information to determine the rate, this option uses incorrect units for a reaction rate, which should be expressed in nM/s, not s/nM.

(Choice C) The units of a reaction rate should be expressed in nM/s, not nM. Additionally, there is not enough information provided to determine the magnitude of the folding rate.

Educational objective:

The rate of a reaction is determined by multiplying the concentrations of the reacting species involved, raised to an empirically determined power, times a rate constant. The rate constant (which is altered by the presence of enzymes or chaperones) must have units such that the rate is expressed as concentration per unit time. The rate can only be calculated if the concentrations of all reactants are known.

Time spent: 0 sec

Subject:
Biochemistry

QID: 401137

System:
Amino Acids and Proteins

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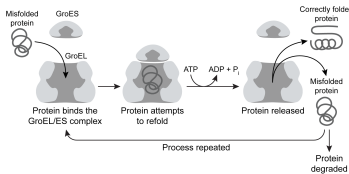


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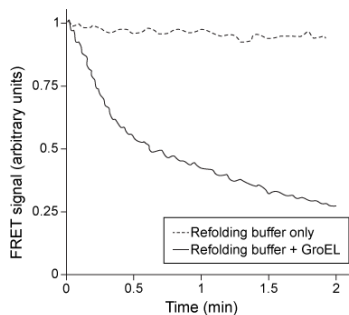


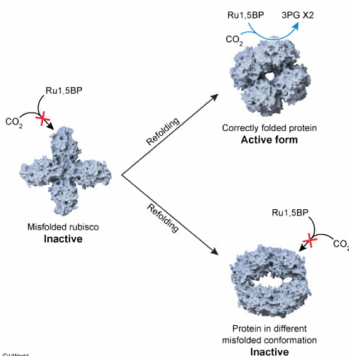
Figure 2 FRET signal of labeled rubisco with and without GroEL

Which of the following experiments, performed in a closed system, could confirm that GroEL/ES successfully facilitated proper rubisco folding?

- ☒ A. Adding Ru1,5BP and CO_2 , and measuring CO_2 consumption over time [50%]
- ☐ B. Running rubisco on an SDS gel to assess its dimerization [27%]
- ☐ C. Observing a decrease in the concentration of inorganic phosphate [8%]
- ☐ D. Measuring the K_d of GroEL/ES interactions with Mg^{2+} over time [13%]

Correct
51% answered correctly

Explanation:



The **function of a protein** is determined by its **three-dimensional folded form**. A properly folded protein will be active, so an assay of its

activity can reveal whether correct folding has been achieved. GroEL/ES is a chaperone protein that, by definition, helps other proteins to fold. Figure 2 shows that rubisco undergoes a conformational change in the presence of GroEL/ES, but it is possible that the form facilitated by GroEL/ES is still not the correctly folded, functional form. To confirm that the functional form has been produced, researchers could **assay rubisco activity**.

The passage states that **rubisco carboxylates its substrate Ru1,5BP** and hydrolyzes it to produce two 3PG molecules as products. Carboxylation is the **attachment of CO₂** to another molecule. In an open system, CO₂ from the atmosphere can dissolve into solution to replace CO₂ consumed by the reaction. However, the question specifies that the experiment is carried out in a closed system, in which matter cannot be exchanged with the environment. In a closed system containing correctly folded rubisco, CO₂, and Ru1,5BP, the concentration of dissolved CO₂ will decrease over time as it is consumed by the reaction. Therefore, **CO₂consumption** under these conditions indicates that **rubisco is properly folded**.

(Choice B) The ability of rubisco to dimerize is an indication that it is folded correctly. However, SDS gels **denature proteins** and destroy tertiary and quaternary structure, including noncovalent dimerization, so they cannot be used to assess proper folding. The passage states that rubisco forms *noncovalent* homodimers, so no disulfide (ie, covalent) bonds are present to hold the monomers bound together in the presence of the denaturant SDS.

(Choice C) GroEL/ES hydrolyzes ATP to ADP and inorganic phosphate (P_i) each time it releases a bound protein, so the P_i concentration *increases* over time. In addition, the passage states that a misfolded protein may require multiple rounds of GroEL/ES binding, and that it may never fold correctly. A change in P_i concentration does not guarantee correct protein folding.

(Choice D) The *K_d* of magnesium binding to *rubisco* may help confirm correct folding, but magnesium binding to GroEL/ES is not related to the conformation of rubisco.

Educational objective:

The function of a protein is tied directly to its native, three-dimensional form. Activity assays can confirm that a protein has folded correctly.

Time spent:0 sec
Subject:
Biochemistry

QID:401138
System:
Amino Acids and Proteins